

Sol: 1(a) - Given a stable LTI system function

A*

1/2

$$H(z) = \frac{1 - \frac{4}{3} \bar{z}^{-1} + \frac{1}{2} \bar{z}^{-2}}{\bar{z}^{-1} (1 - \frac{1}{2} \bar{z}^{-1}) (1 - \frac{1}{3} \bar{z}^{-1})} \quad - (1)$$

poles at
 $\bar{z}^{-1} = 0 \Rightarrow z = \infty$
 $1 - \frac{1}{2} \bar{z}^{-1} = 0 \Rightarrow z = \frac{1}{2}$
 $1 - \frac{1}{3} \bar{z}^{-1} = 0 \Rightarrow z = \frac{1}{3}$

By Partial fraction, we need to simplify $H(z)$

$$\text{So, } H(z) = \frac{A}{\bar{z}^{-1}} + \frac{B}{1 - \frac{1}{2} \bar{z}^{-1}} + \frac{C}{1 - \frac{1}{3} \bar{z}^{-1}}$$

$$\Rightarrow \frac{1 - \frac{4}{3} \bar{z}^{-1} + \frac{1}{2} \bar{z}^{-2}}{\bar{z}^{-1} (1 - \frac{1}{2} \bar{z}^{-1}) (1 - \frac{1}{3} \bar{z}^{-1})} = \frac{A}{\bar{z}^{-1}} + \frac{B}{1 - \frac{1}{2} \bar{z}^{-1}} + \frac{C}{1 - \frac{1}{3} \bar{z}^{-1}}$$

$$\Rightarrow 1 - \frac{4}{3} \bar{z}^{-1} + \frac{1}{2} \bar{z}^{-2} = A(1 - \frac{1}{2} \bar{z}^{-1})(1 - \frac{1}{3} \bar{z}^{-1}) + B(\bar{z}^{-1})(1 - \frac{1}{3} \bar{z}^{-1}) + C(\bar{z}^{-1})(1 - \frac{1}{2} \bar{z}^{-1}) \quad - (2)$$

Now Put $\bar{z}^{-1} = 0$, we get

$$\Rightarrow 1 - \frac{4}{3} \times 0 + \frac{1}{2} (0) = A(1 - \frac{1}{2} \times 0)(1 - \frac{1}{3} (0)) + B \times 0 (1 - \frac{1}{3} \bar{z}^{-1}) + C \times 0 (1 - \frac{1}{2} \bar{z}^{-1})$$

$$\Rightarrow 1 = A \times 1 \times 1$$

$$\text{So, } \boxed{A = 1}$$

Now Put $\bar{z}^{-1} = 2$, we get

$$\Rightarrow 1 - \frac{4}{3} \times 2 + \frac{1}{2} \times 2 \times 2 = A(1 - \frac{1}{2} \times 2)(1 - \frac{1}{3} \times 2) + B \times 2 (1 - \frac{1}{3} \times 2) + C \times 2 (1 - \frac{1}{2} \times 2)$$

$$\Rightarrow 1 - \frac{8}{3} + 2 = A(1-1)(1 - \frac{2}{3}) + 2B(1 - \frac{2}{3}) + C \times 2(1-1)$$

$$\Rightarrow 3 - \frac{8}{3} = A \times 0 + 2B \times (\frac{3-2}{3}) + C \times 2 \times 0$$

$$\Rightarrow \frac{9-8}{3} = 2B \times 1/3$$

$$\Rightarrow 1/3 = 2B \times 1/3$$

$$\text{So } \boxed{B = 1/2}$$

Now put $\bar{z}^1 = 3$, we get

$$\Rightarrow 1 - \frac{4}{3} \times 3 + \frac{1}{2} \times 3 \times 3 = A \left(1 - \frac{1}{2} \times 3\right) \left(1 - \frac{1}{3} \times 3\right) + B \times 3 \left(1 - \frac{1}{3} \times 3\right) + C \times 3 \left(1 - \frac{1}{2} \times 3\right)$$

$$\Rightarrow 1 - 4 + \frac{9}{2} = A \left(1 - 3/2\right) \times 0 + 3 \times B \times 0 + C \times 3 \times \left(1 - \frac{3}{2}\right)$$

$$\Rightarrow -3 + \frac{9}{2} = 3C \times \left(\frac{2-3}{2}\right)$$

$$\Rightarrow \frac{-6+9}{2} = \frac{3}{2} = 3C \times \left(-1/2\right)$$

$$\Rightarrow \frac{3}{2} = -3 \times \frac{C}{2}$$

$$\text{So, } \boxed{C = -1}$$

Now put the value of A, B & C in equation ①, we get

$$H(z) = \frac{1}{\bar{z}^1} + \frac{1/2}{1 - \frac{1}{2} \bar{z}^1} - \frac{1}{1 - \frac{1}{3} \bar{z}^1}$$

From Ath Poles

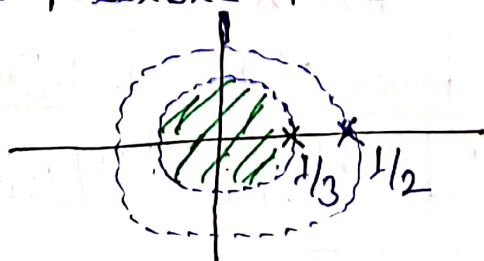
$$z = \infty, z = 1/2$$

$$\& z = 1/3$$

Region of Convergence

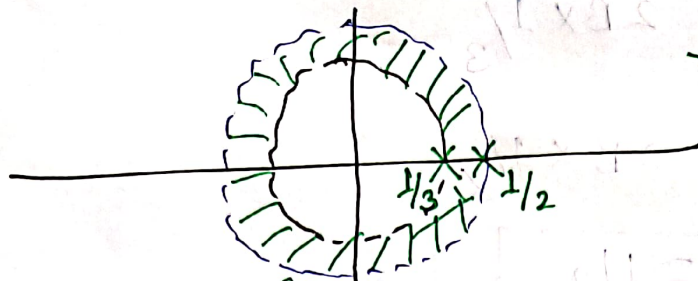
→ There are three possible ROC

①



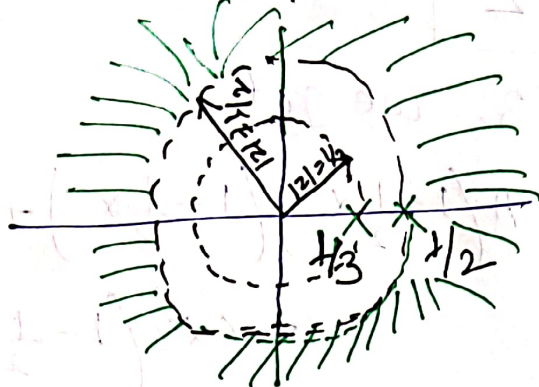
$$\boxed{0 < |z| < 1/3}$$

⑥



$$\boxed{1/3 < |z| < 1/2}$$

⑦



$$\boxed{|z| > 1/2 \text{ but excludes } |z| = \infty}$$

Since it is given that the LTI system is stable, the ROC should include unit circle i.e. $|z|=1$. Hence it implies that the overall ROC is $|z| > 1/2$ but excludes $|z| = \infty$.

→ ① mark

Impulse Response $h[n]$

$$\rightarrow \boxed{X(z) \triangleq \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n}} \quad x[n] \xleftrightarrow{z} X(z)$$

We know that

$$\begin{aligned} \rightarrow & \alpha^n u[n] \xleftrightarrow{z} \frac{1}{1 - \alpha z^{-1}} \quad |z| > |\alpha| \quad \text{--- ① mark} \\ \rightarrow & x[n - n_0] \xleftrightarrow{z} z^{-n_0} X(z) \text{ with ROC } = R, \text{ except for the possible addition \& deletion of the origin or} \\ \rightarrow & \delta[n] \xleftrightarrow{z} 1 \\ \rightarrow & \delta[n - 1] \xleftrightarrow{z} \sum_{n=-\infty}^{\infty} \delta[n - 1] \cdot z^{-n} = z^{-1} \text{ infinity} \\ \rightarrow & \boxed{\delta[n + 1] \xleftrightarrow{z} \sum_{n=-\infty}^{\infty} \delta[n + 1] \cdot z^{-n} = z} \end{aligned}$$

Now the impulse response of $H(z)$ is $h[n]$

$$\text{so, } \boxed{h[n] = \delta[n + 1] + \frac{1}{2} \cdot \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{3}\right)^n u[n]} \quad \text{--- ① mark}$$

Causality

4

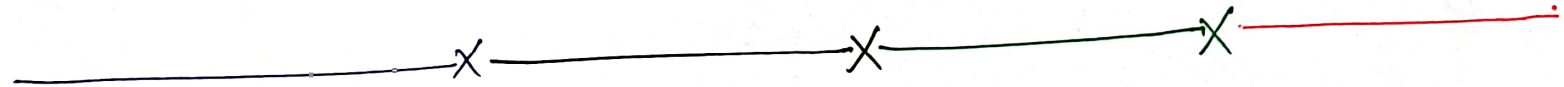
→ A discrete-time LTI system is causal if & only if the ROC of its system function is the exterior of a circle, including infinity.

→ Here the ROC is the exterior of a circle (i.e. $|z| > 1/2$) but it excludes the infinity. so this is not Causal.

or,

→ Here one pole is sitting at $z = \infty$, so this system is not Causal.

→ ① mark



Sol: 1(b) Given stable LTI system

$$H(z) = \frac{z - 1/2}{z^2 + \frac{z}{2} - \frac{3}{16}} = \frac{P(z)}{Q(z)} \quad \text{--- (1)}$$

poles
when

$$H(z) = \frac{N(z)}{D(z)}$$

$$D(z) \neq 0$$

& hence;

Here $Q(z) = z^2 + \frac{z}{2} - \frac{3}{16}$, this is Quadratic Polynomial in the form of $P(x) = ax^2 + bx + c = (x + \alpha)(x + \beta)$

Here $a = 1, b = 1/2$ & $c = -\frac{3}{16}$

we know that from Quadratic Polynomial the solution of x is α & β . i.e. root are α & β .

$$\alpha = \frac{-b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a} \quad \& \quad \beta = \frac{-b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a}$$

Here $D = \sqrt{b^2 - 4ac}$

$$= \sqrt{\left(\frac{1}{2}\right)^2 - 4 \times 1 \times \left(-\frac{3}{16}\right)}$$

$$= \sqrt{\frac{1}{4} + \frac{3}{4}}$$

$$= \sqrt{\frac{1}{4} + \frac{3}{4}}$$

$$D = \sqrt{\frac{4}{4}} = \sqrt{1} = 1$$

So, $\alpha = \frac{-b}{2a} + \frac{D}{2a}$

$$= \frac{-1/2}{2 \times 1} + \frac{1}{2 \times 1}$$

$$= -\frac{1}{4} + \frac{1}{2}$$

$$\alpha = \frac{-1 + 2}{4} = \frac{1}{4}$$

$$\beta = \frac{-b}{2a} - \frac{D}{2a} \quad (6)$$

$$= \frac{-1/2}{2 \times 1} - \frac{1}{2 \times 1}$$

$$= -\frac{1}{4} - \frac{1}{2}$$

$$= \frac{-1-2}{4}$$

$$\boxed{\beta = -\frac{3}{4}}$$

$$\text{So, } Q(z) = z^2 + \frac{z}{2} - \frac{3}{16} = \cancel{\left(z - \frac{1}{4}\right)} \cancel{\left(z + \frac{3}{4}\right)}$$

$$\boxed{Q(z) = \left(z - \frac{1}{4}\right) \left(z + \frac{3}{4}\right)}$$

$$\text{So, } H(z) = \frac{z - \frac{1}{2}}{\left(z - \frac{1}{4}\right) \left(z + \frac{3}{4}\right)}$$

$\Rightarrow$

Given poles

at $z = 1/4$

& $z = -3/4$

By partial fraction

$$H(z) = \frac{z - 1/2}{\left(z - \frac{1}{4}\right) \left(z + \frac{3}{4}\right)} = \frac{A}{z - \frac{1}{4}} + \frac{B}{z + \frac{3}{4}}$$

$$\Rightarrow z - 1/2 = A \left(z + \frac{3}{4}\right) + B \left(z - \frac{1}{4}\right)$$

Put $z = 1/4$

$$\Rightarrow \frac{1}{4} - \frac{1}{2} = A \left(\frac{1}{4} + \frac{3}{4}\right) + B \left(\frac{1}{4} - \frac{1}{4}\right)$$

$$\Rightarrow \frac{1-2}{4} = A \times \left(\frac{4}{4}\right)$$

$$\Rightarrow \frac{-1}{4} = A$$

$$\boxed{A = -\frac{1}{4}}$$

Put $z = -\frac{3}{4}$

$$\Rightarrow -\frac{3}{4} - \frac{1}{2} = A\left(-\frac{3}{4} + \frac{3}{4}\right) + B\left(-\frac{3}{4} - \frac{1}{4}\right)$$

$$\Rightarrow \frac{-3-2}{4} = A \times 0 + B\left(\frac{-3-1}{4}\right)$$

$$\Rightarrow \frac{-5}{4} = B \times \frac{-4}{4}$$

$$\Rightarrow -B = -\frac{5}{4}$$

So, $B = \frac{5}{4}$

Now, put the value of A & B in equation (2), we get

$$H(z) = \frac{-1/4}{z - 1/4} + \frac{5/4}{z + \frac{3}{4}} \quad (3)$$

$$H(z) = H_1(z) + H_2(z) \quad (4)$$

Here, $H_1(z) = \frac{-1/4}{z - 1/4}$

Divide Numerator & Denominator by $\frac{1}{2}z$, we get

$$H_1(z) = \frac{(-1/4) \times \frac{1}{2}}{\frac{1}{2}\left(z - \frac{1}{4}\right)}$$

$$H_1(z) = \frac{-\frac{1}{4} z^{-1}}{1 - \frac{1}{4} z^{-1}} \quad (5)$$

$$H_2(z) = \frac{5/4}{z + 3/4}$$

→ Divide Numerator & Denominator by z , we get

$$H_2(z) = \frac{5/4 \bar{z}^{-1}}{1 + 3/4 \bar{z}^{-1}} \quad - (6)$$

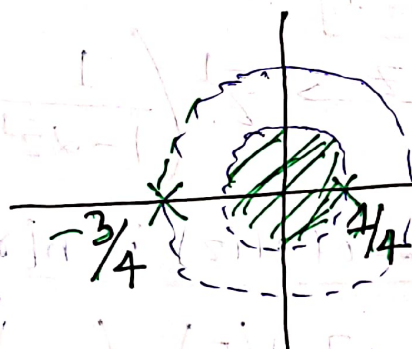
Putting $H_1(z)$ & $H_2(z)$ Value in equation (4), we get

$$H(z) = \frac{-1/4 \bar{z}^{-1}}{1 - 1/4 \bar{z}^{-1}} + \frac{5/4 \bar{z}^{-1}}{1 + 3/4 \bar{z}^{-1}} \quad - (7)$$

Region of Convergence

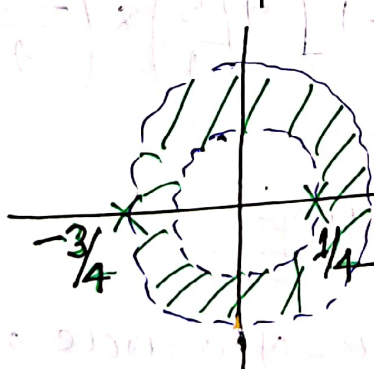
→ There are three possible ROC

(a)



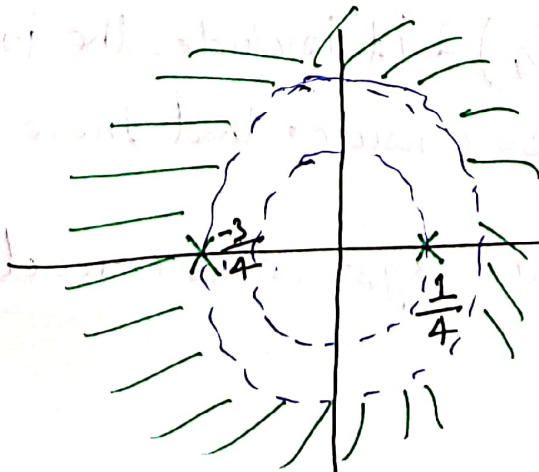
$$0 < |z| < 1/4$$

(b)



$$\frac{1}{4} < |z| < \frac{3}{4}$$

(c)



$$|z| > \frac{3}{4}$$

→ since it is given that the LTI system is stable, the ROC should include unit circle i.e. $|z|=1$. Hence it implies that the overall ROC is $|z| > \frac{3}{4}$ — option C. (9)

→ 1 mark

Impulse Response $h[n]$

We know that

→ $x[n-n_0] \xleftrightarrow{Z} \bar{z}^{n_0} X(z)$ with ROC = R, except for the possible addition or deletion of the origin & infinity

→ $\alpha^n u[n] \xleftrightarrow{Z} \frac{1}{1-\alpha \bar{z}^{-1}} \quad |z| > |\alpha|$

→ $\alpha^{n-n_0} \cdot u[n-n_0] \xleftrightarrow{Z} \bar{z}^{n_0} \left(\frac{1}{1-\alpha \bar{z}^{-1}} \right)$

[Put $n_0=1$]

→ $\alpha^{n-1} \cdot u[n-1] \xleftrightarrow{Z} \bar{z}^{-1} \left(\frac{1}{1-\alpha \bar{z}^{-1}} \right)$

Now the impulse response of $H(z)$ is $h[n]$ (1)

$$h[n] = \left(-\frac{1}{4}\right) \times \left(\frac{1}{4}\right)^{n-1} \cdot u[n-1] + \left(\frac{5}{4}\right) \times \left(-\frac{3}{4}\right)^{n-1} \cdot u[n-1]$$

Causality

→ Here the ROC is the exterior of a circle outside the outermost pole ($|z| > \frac{3}{4}$) & it includes the infinity. This impulse response also ensures that there is no pole sitting at $z = \infty$.

So this system is Causal.

→ 1 mark